Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*		$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{3.8}$ day^{-1} [=0.182 day^{-1}] (1) Activity after 12 days $A = A_0 \exp\left(-\frac{12\ln 2}{3.8}\right)$(1) substitution = $0.112A_0$ (1) $\therefore$ % reduction = 88.8%(1) (Accept variation because of rounding) **Alternative** Number of half-lives, $n = \frac{12}{3.8} = 3.16$ [or by implication] (1) Fraction after 12 days $= 2^{-3.16}$ (1) = 0.112 (1) = 11.2% $\therefore$ Percentage reduction = 88.8% (1)		4		4	4	

	(b)		Any 4 × (1) 1) Counts is reduced significantly (or equivalent alternatives e.g. by almost a half) by the paper, so alpha particles present (✓) 2) Another significant reduction (or alternative e.g. essentially all of the remaining radiation is stopped) by the aluminium, so beta particles present (✓) 3) Count with lead is larger than with aluminium (but almost the same) so no gamma present (✓) 4) The measured counts with aluminium and lead are essentially the same, so this is because of background radiation / the background radiation is approximately 25 counts per minute. (✓) 5) Randomness of nuclear decay is the reason for increased value with lead. (✓)		4		4		4
			Question 5 total	**0**	**8**	**0**	**8**	**4**	**4**